1.5 Multivariable Calculus: Solutions

These are solutions to problems.md. The numerical answers (Problems 1, 6, 9, 13, 14, 19, 20b) were checked with a computer algebra system.

Warm-up

1. $\nabla f = (2xy,\ x^2 + ze^{yz},\ ye^{yz})$, so $\nabla f(1,1,0) = (2,1,1)$.

The unit vector in the given direction is $u = (1,2,2)/3$, since $|(1,2,2)| = 3$. So

$$D_u f(1,1,0) = (2,1,1) \cdot \tfrac{1}{3}(1,2,2) = \tfrac{2+2+2}{3} = 2.$$

2.

$$JF = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det JF = r\cos^2\theta + r\sin^2\theta = r.$$

$DF(r,\theta)$ is invertible iff $r \neq 0$. At $r = 0$, every θ maps to the origin, so polar coordinates degenerate there.

3. $f_x = 2x = 2$ and $f_y = 2y = 4$ at $(1,2)$. The tangent plane is

$$z = 5 + 2(x-1) + 4(y-2), \quad \text{i.e.} \quad 2x + 4y - z = 5.$$

For $G = x^2 + y^2 - z$, $\nabla G = (2x, 2y, -1) = (2,4,-1)$ at $(1,2,5)$. This is the normal of the plane $2x + 4y - z = 5$, so the two answers agree. The point lies on the plane: $2 + 8 - 5 = 5$.

4. By the chain rule, with $\partial x/\partial r = \cos\theta$, $\partial y/\partial r = \sin\theta$, $\partial x/\partial\theta = -r\sin\theta$ and $\partial y/\partial\theta = r\cos\theta$:

$$z_r = z_x\cos\theta + z_y\sin\theta, \qquad z_\theta = -z_x r\sin\theta + z_y r\cos\theta.$$

Write $c = \cos\theta$ and $s = \sin\theta$. Then

$$z_r^2 + \tfrac{1}{r^2}z_\theta^2 = (cz_x + sz_y)^2 + (-sz_x + cz_y)^2 = (c^2+s^2)z_x^2 + (s^2+c^2)z_y^2 + (2cs - 2cs)z_x z_y = z_x^2 + z_y^2.$$

(Geometrically, $(z_r,\ z_\theta/r)$ is ∇z written in the rotated orthonormal frame $(\hat r, \hat\theta)$, so it has the same length.)

5.

$$\int_0^1 \int_0^2 (x+y)\,dy\,dx = \int_0^1 \left[xy + \tfrac{y^2}{2}\right]_0^2 dx = \int_0^1 (2x+2)\,dx = 1 + 2 = 3.$$

6. The region is $\{0 \le x \le 1,\ x \le y \le 1\} = \{0 \le y \le 1,\ 0 \le x \le y\}$, a triangle. In the reversed order:

$$\int_0^1 \int_0^y e^{y^2}\,dx\,dy = \int_0^1 ye^{y^2}\,dy = \left[\tfrac{1}{2}e^{y^2}\right]_0^1 = \frac{e-1}{2}.$$

The original order cannot be done directly, because e^{y^2} has no elementary antiderivative.

Core

7. (a) $f(x,0) = 0$ for all x, so $f_x(0,0) = \lim_{t\to 0}\frac{f(t,0)-0}{t} = 0$. Similarly $f_y(0,0) = 0$. But on the line $y = x$, $f(t,t) = \frac{t^2}{2t^2} = \frac{1}{2}$ for $t \neq 0$. So $f(t,t) \to \frac{1}{2} \neq 0 = f(0,0)$, and f is not continuous at 0, let alone differentiable.

(b) For a unit vector $u = (u_1, u_2)$ and $t \neq 0$:

$$\frac{g(tu) - g(0)}{t} = \frac{t^3 u_1^2 u_2}{t^2(u_1^2 + u_2^2)\, t} = u_1^2 u_2,$$

so $D_u g(0) = u_1^2 u_2$ exists for every u.

If g were differentiable at 0, we would have $D_u g(0) = \nabla g(0) \cdot u$. Now $\nabla g(0) = (D_{e_1} g(0), D_{e_2} g(0)) = (0, 0)$, so every directional derivative would be 0. But $u = (1, 1)/\sqrt{2}$ gives $D_u g(0) = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \neq 0$. So g is not differentiable at 0. (In fact g is continuous: $|g| \leq |y|$.)

8. Let $a = (a_1, a_2)$ and $h = (h_1, h_2)$, small enough that the partials exist on the rectangle spanned. Split the increment into two one-variable steps:

$$f(a + h) - f(a) = \big[f(a_1 + h_1, a_2 + h_2) - f(a_1, a_2 + h_2)\big] + \big[f(a_1, a_2 + h_2) - f(a_1, a_2)\big].$$

Apply the one-variable mean value theorem to each bracket. The first is a function of x alone, and f_x exists near a; the second is a function of y alone. This gives

$$f(a + h) - f(a) = f_x(\xi) h_1 + f_y(\eta) h_2,$$

with $\xi = (a_1 + \theta_1 h_1,\ a_2 + h_2)$ and $\eta = (a_1,\ a_2 + \theta_2 h_2)$ for some $\theta_i \in (0, 1)$. Hence

$$\frac{|f(a + h) - f(a) - f_x(a) h_1 - f_y(a) h_2|}{|h|} \leq |f_x(\xi) - f_x(a)| + |f_y(\eta) - f_y(a)|,$$

using $|h_i| \leq |h|$. As $h \to 0$, both $\xi \to a$ and $\eta \to a$. By continuity of the partials at a, the right side tends to 0. So f is differentiable at a, with $Df(a) = (f_x(a), f_y(a))$.

9. $\nabla f = (3x^2 - 3y,\ 3y^2 - 3x) = 0$ gives $y = x^2$ and $x = y^2$. So $x = x^4$, i.e. $x(x^3 - 1) = 0$, and $x = 0$ or $x = 1$ (the only real roots). The critical points are $(0, 0)$ and $(1, 1)$.

The Hessian is $Hf = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$.

- At $(0, 0)$: $D = \det Hf = 0 - 9 = -9 < 0$, so it is a **saddle**.
- At $(1, 1)$: $D = 36 - 9 = 27 > 0$ and $f_{xx} = 6 > 0$, so it is a **local minimum**, with $f(1, 1) = -1$.

Neither is a global extremum, because $f(x, 0) = x^3$ is unbounded in both directions.

10. Let $g = x^2 + y^2 + z^2$. Then $\nabla g = 2(x, y, z) \neq 0$ on the sphere, and the sphere is compact, so a maximum and minimum exist and occur at Lagrange points.

$\nabla f = (1, 2, 2) = \lambda \cdot 2(x, y, z)$ forces $\lambda \neq 0$ and $(x, y, z) = \frac{1}{2\lambda}(1, 2, 2)$. Then $|(x, y, z)| = 3$ gives $\frac{3}{2|\lambda|} = 3$, so $\lambda = \pm\frac{1}{2}$ and $(x, y, z) = \pm(1, 2, 2)$.

- $f(1, 2, 2) = 9$ is the **maximum**.
- $f(-1, -2, -2) = -9$ is the **minimum**.

Check: by Cauchy–Schwarz, $|f| = |(1, 2, 2) \cdot (x, y, z)| \leq 3 \cdot 3 = 9$, with equality iff (x, y, z) is parallel to $(1, 2, 2)$.

11. We extremise $f = z$ subject to $g = x^2 + y^2 = 1$ and $h = x + y + z = 1$. On the curve, $\nabla g = (2x, 2y, 0) \neq 0$ and $\nabla h = (1, 1, 1)$, and these are independent because ∇g is nonzero with zero z-component. The curve is compact (an ellipse).

$\nabla f = \lambda \nabla g + \mu \nabla h$ reads

$$(0, 0, 1) = \lambda(2x, 2y, 0) + \mu(1, 1, 1).$$

- The third component gives $\mu = 1$.
- Then $2\lambda x = -1$ and $2\lambda y = -1$. So $\lambda \neq 0$ and $x = y = -\frac{1}{2\lambda}$.
- From $x^2 + y^2 = 1$: $x = y = \pm\frac{1}{\sqrt{2}}$, and $z = 1 - x - y = 1 \mp \sqrt{2}$.

So the **highest point** is $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 1+\sqrt{2}\right)$ and the **lowest point** is $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1-\sqrt{2}\right)$.

Check: on the circle, $x + y = \sqrt{2}\cos(t - \pi/4)$ ranges over $[-\sqrt{2}, \sqrt{2}]$.

12. The integrand is positive, so the improper integral may be computed in any coordinates as a limit over discs (Tonelli; made rigorous in 3.1):

$$\iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dA = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta = 2\pi\left[-\tfrac{1}{2}e^{-r^2}\right]_0^{\infty} = \pi.$$

By Fubini, the same integral equals $\left(\int_{\mathbb{R}} e^{-x^2}dx\right)^2$. The integral is positive, so $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$.

13. In spherical coordinates, the cone $z = \sqrt{x^2+y^2}$ is $\varphi = \pi/4$ and the sphere is $\rho = 2$. The region is $0 \le \rho \le 2$, $0 \le \varphi \le \pi/4$, $0 \le \theta \le 2\pi$. So

$$V = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{2} \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta = 2\pi \cdot \left(1 - \tfrac{\sqrt{2}}{2}\right) \cdot \tfrac{8}{3} = \frac{8\pi(2-\sqrt{2})}{3} \approx 4.91.$$

14. Let $u = x + y$ and $v = x - y$. Then D becomes the rectangle $0 \le u \le 1$, $0 \le v \le 2$, and

$$\frac{\partial(u,v)}{\partial(x,y)} = \det\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2, \quad \text{so} \quad dA_{xy} = \tfrac{1}{2}\,du\,dv.$$

Hence

$$\iint_D (x+y)e^{x-y}dA = \int_0^1\int_0^2 ue^v \cdot \tfrac{1}{2}\,dv\,du = \tfrac{1}{2}\cdot\tfrac{1}{2}\cdot(e^2-1) = \frac{e^2-1}{4}.$$

15. (a) $P = y$ and $Q = x$, so $Q_x = 1 = P_y$ on all of $\mathbb{R}^2$, which is simply connected. Hence F is conservative. A potential is $\varphi = xy$: $\varphi_x = y$ and $\varphi_y = x$. For any curve C from $(0,0)$ to $(1,2)$, $\int_C F \cdot dr = \varphi(1,2) - \varphi(0,0) = 2$.

(b) With $P = \frac{-y}{x^2+y^2}$ and $Q = \frac{x}{x^2+y^2}$:

$$P_y = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}, \qquad Q_x = \frac{(x^2+y^2)-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

So $Q_x = P_y$.

On the unit circle $r(t) = (\cos t, \sin t)$, $G(r(t)) = (-\sin t, \cos t) = r'(t)$. So $G \cdot r' = 1$ and $\oint G \cdot dr = \int_0^{2\pi} 1\,dt = 2\pi$.

No contradiction: "curl-free ⇒ conservative" requires a simply connected domain, and $\mathbb{R}^2 \setminus \{0\}$ is not simply connected. Equivalently, Green's theorem cannot be applied to the unit disc, because G is not defined (let alone C^1) at 0. Locally $G = \nabla\theta$, but θ has no continuous single-valued branch around the origin.

16. Parametrise the ellipse by $x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$ (counter-clockwise). Then

$$\text{Area} = \frac{1}{2}\oint x\,dy - y\,dx = \frac{1}{2}\int_0^{2\pi} (a\cos t \cdot b\cos t + b\sin t \cdot a\sin t)dt = \frac{1}{2}\cdot ab\cdot 2\pi = \pi ab.$$

17. (a) On $|x| = R$, the outward unit normal is $n = x/R$, so $F \cdot n = x \cdot x/R = R^2/R = R$ is constant. The flux is $R \cdot \text{Area} = R \cdot 4\pi R^2 = 4\pi R^3$.

(b) $\operatorname{div} F = 1+1+1 = 3$. The flux is $\iiint_{|x|\le R} 3\,dV = 3 \cdot \frac{4}{3}\pi R^3 = 4\pi R^3$. The two answers agree.

18. *Line integral.* With the upward normal, the right-hand rule orients ∂S (the unit circle in $z = 0$) counter-clockwise when viewed from above: $r(t) = (\cos t, \sin t, 0)$. Then $F(r(t)) = (-\sin t, \cos t, 0)$ and $F \cdot r' = \sin^2 t + \cos^2 t = 1$. So $\oint_{\partial S} F \cdot dr = 2\pi$.

Surface integral. $\operatorname{curl} F = (0-0,\ 0-0,\ 1-(-1)) = (0,0,2)$. On the unit sphere, $n = (x,y,z)$, so $\operatorname{curl} F \cdot n = 2z = 2\cos\varphi$. With $dS = \sin\varphi\, d\varphi\, d\theta$:

$$\iint_S 2\cos\varphi\, dS = \int_0^{2\pi}\int_0^{\pi/2} 2\cos\varphi\sin\varphi\, d\varphi\, d\theta = 2\pi \cdot 2 \cdot \tfrac{1}{2} = 2\pi.$$

Both sides equal 2π. (Shortcut: the flux of $(0,0,2)$ through any surface with this boundary equals its flux through the flat unit disc, which is $2 \cdot \pi$.)

19. $\operatorname{div} F = 3(x^2+y^2+z^2) = 3\rho^2$. By the divergence theorem:

$$\iiint_{\rho\le 1} 3\rho^2\, dV = \int_0^{2\pi}\int_0^{\pi}\int_0^{1} 3\rho^4 \sin\varphi\, d\rho\, d\varphi\, d\theta = 3 \cdot \tfrac{1}{5} \cdot 2 \cdot 2\pi = \frac{12\pi}{5}.$$

Challenge

20. (a) Fix (a,b) and small $h \neq 0$, and set

$$\Delta(h) = f(a+h, b+h) - f(a+h, b) - f(a, b+h) + f(a,b).$$

First grouping. Let $\varphi(x) = f(x, b+h) - f(x,b)$, so $\Delta = \varphi(a+h) - \varphi(a)$. By the MVT, $\Delta = h\varphi'(\xi_1) = h\big[f_x(\xi_1, b+h) - f_x(\xi_1, b)\big]$ for some ξ_1 between a and $a+h$. Applying the MVT again to $y \mapsto f_x(\xi_1, y)$ gives $\Delta = h^2 (f_x)_y(\xi_1, \eta_1)$, with η_1 between b and $b+h$.

Second grouping. Let $\psi(y) = f(a+h, y) - f(a,y)$, so $\Delta = \psi(b+h) - \psi(b)$. The same two steps give $\Delta = h^2 (f_y)_x(\xi_2, \eta_2)$.

So $(f_x)_y(\xi_1,\eta_1) = (f_y)_x(\xi_2,\eta_2)$, where both points lie in the square of side $|h|$ at (a,b). Let $h \to 0$. By continuity of the second partials, $(f_x)_y(a,b) = (f_y)_x(a,b)$.

(b) *Compute* $f_x(0,y)$. For $y \neq 0$:

$$f_x(0,y) = \lim_{t\to 0} \frac{f(t,y) - f(0,y)}{t} = \lim_{t\to 0} \frac{y(t^2-y^2)}{t^2+y^2} = \frac{y(-y^2)}{y^2} = -y.$$

Also $f_x(0,0) = 0$, since $f(t,0) = 0$. So $f_x(0,y) = -y$ for all y, and $\frac{\partial}{\partial y} f_x(0,0) = -1$.

Compute $f_y(x,0)$. For $x \neq 0$:

$$f_y(x,0) = \lim_{t\to 0} \frac{f(x,t)}{t} = \frac{x \cdot x^2}{x^2} = x.$$

Also $f_y(0,0) = 0$. So $\frac{\partial}{\partial x} f_y(0,0) = 1$.

The mixed partials are $-1 \neq 1$. This is consistent with (a), because the second partials of f are not continuous at 0.

21. *At I.* By the Leibniz formula, $\det(I+H) = \sum_\sigma \operatorname{sgn}\sigma \prod_i (\delta_{\sigma(i)i} + H_{\sigma(i)i})$.

- The identity permutation contributes $\prod_i(1+H_{ii}) = 1 + \sum_i H_{ii} +$ (terms of degree ≥ 2 in the entries of H).
- Every other σ moves at least two indices, so its product contains at least two off-diagonal entries of H. Its degree is at least 2.

So $\det(I+H) = 1 + \operatorname{tr} H + R(H)$, where R is a polynomial whose terms all have degree ≥ 2. Hence $|R(H)| \leq C|H|^2$ for $|H| \leq 1$, and $|R(H)|/|H| \to 0$. So $D\det(I)(H) = \operatorname{tr} H$.

At invertible A. $\det(A+H) = \det A \cdot \det(I + A^{-1}H) = \det A\,(1 + \operatorname{tr}(A^{-1}H) + R(A^{-1}H))$. Since $|A^{-1}H| \leq \|A^{-1}\|\,|H|$, the remainder is $O(|H|^2)$. So

$$D\det(A)(H) = \det A \cdot \operatorname{tr}(A^{-1}H).$$

(Equivalently, $\operatorname{tr}(\operatorname{adj}(A)H)$, which extends to all A.)

22. (a) $q(x) = \sum_{i,j} a_{ij}x_ix_j$, so

$$\partial_k q = \sum_j a_{kj}x_j + \sum_i a_{ik}x_i = 2(Ax)_k,$$

using $a_{ik} = a_{ki}$. So $\nabla q = 2Ax$.

(b) The unit sphere $S = \{g(x) = |x|^2 = 1\}$ is compact and q is continuous, so q attains its maximum on S at some x_0. Since $\nabla g = 2x \neq 0$ on S, Lagrange gives $2Ax_0 = \lambda \cdot 2x_0$. So $x_0 \neq 0$ is an eigenvector with **real** eigenvalue λ, and $q(x_0) = x_0^t A x_0 = \lambda |x_0|^2 = \lambda$.

If μ is any real eigenvalue with unit eigenvector v, then $\mu = q(v) \leq q(x_0) = \lambda$. Combined with the spectral theorem, under which all eigenvalues are real, $\max_{|x|=1} q = \lambda_{\max}$.

(This gives the base case of an induction proof of the real spectral theorem: restrict to $x_0^\perp$ and repeat.)

23. (a) Let $r = |x|$, so $\partial_i r = x_i/r$. Then

$$\partial_i\left(\frac{x_i}{r^3}\right) = \frac{1}{r^3} - \frac{3x_i^2}{r^5}, \qquad \operatorname{div} F = \frac{3}{r^3} - \frac{3r^2}{r^5} = 0 \quad (x \neq 0).$$

(b) On $|x| = R$, $n = x/R$, so $F \cdot n = \frac{x\cdot x}{R^3 \cdot R} = \frac{1}{R^2}$. The flux is $\frac{1}{R^2} \cdot 4\pi R^2 = 4\pi$, independent of R.

(c) The divergence theorem on the ball $|x| \leq R$ needs F to be C^1 on the whole ball. F is undefined at 0, so the theorem does not apply, and there is no contradiction.

Now let $S = \partial E$, where E is a nice solid with 0 in its interior. Choose $\varepsilon > 0$ with $\overline{B_\varepsilon(0)} \subset \operatorname{int} E$. Let $E_\varepsilon = E \setminus B_\varepsilon(0)$. F is C^1 on E_ε, whose boundary is S (outward normal) together with the small sphere $|x| = \varepsilon$, whose outward normal for E_ε points *towards* 0. The divergence theorem on E_ε gives

$$0 = \iiint_{E_\varepsilon} \operatorname{div} F\, dV = \iint_S F \cdot n\, dS - \iint_{|x|=\varepsilon} F \cdot \frac{x}{\varepsilon}\, dS = \iint_S F \cdot n\, dS - 4\pi.$$

So the flux through S is 4π. If instead $0 \notin \overline{E}$, the flux is 0. This is Gauss's law: up to constants, the flux counts the enclosed charge.